

CRYPTOGRAPHY

2 Types of cryptography

Encryption

symmetric

1 key (decryption and encryption)

asymmetric

2 keys

1 public key (encryption)

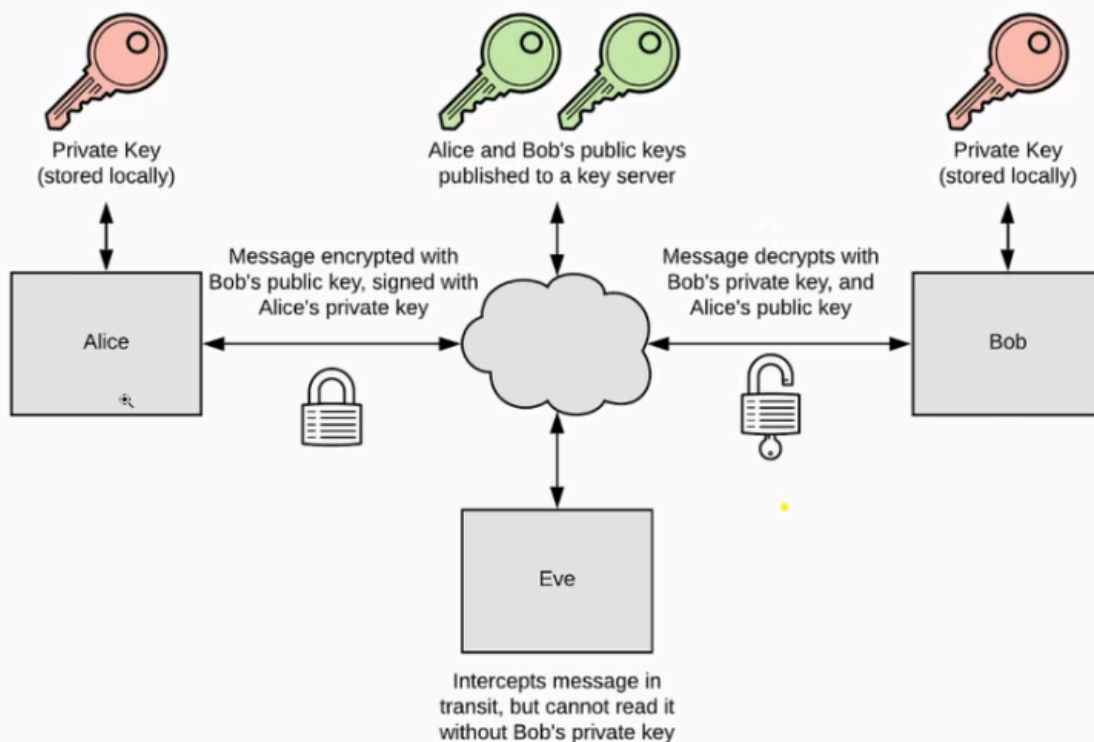
1 private key (decryption)

Hash function

A hash function is a mathematical algorithm that takes an input of any size

The Hash function is:

- Deterministic
- Fixed Length
- a one way function



RSA algorithm:

Key Generation	
Select p, q	p and q both prime
Calculate $n = p \times q$	
Calculate $\phi(n) = (p-1)(q-1)$	
Select integer e	$\gcd(\phi(n), e) = 1; 1 < e < \phi(n)$
Calculate d	$d \equiv e^{-1} \pmod{\phi(n)}$
Public key	$KU = \{e, n\}$
Private key	$KR = \{d, n\}$

Encryption	
Plaintext	$M < n$
Ciphertext	$C = M^e \pmod{n}$

Decryption	
Ciphertext	C
Plaintext	$M = C^d \pmod{n}$

Virus total

<https://www.virustotal.com/gui/home/upload>

Give him the Hash of something u want to download

U crack hash, u don't decrypt

symmetric encryption examples:

- Advanced Encryption Standard (AES)
- ChaCha20
- Data Encryption Standard (DES)

DES brute forceable

asymmetric encryption examples:

- RSA (Rivest-Shamir-Adleman)
- ECC (Elliptic Curve Cryptography)
- Diffie-Hellman (DH)

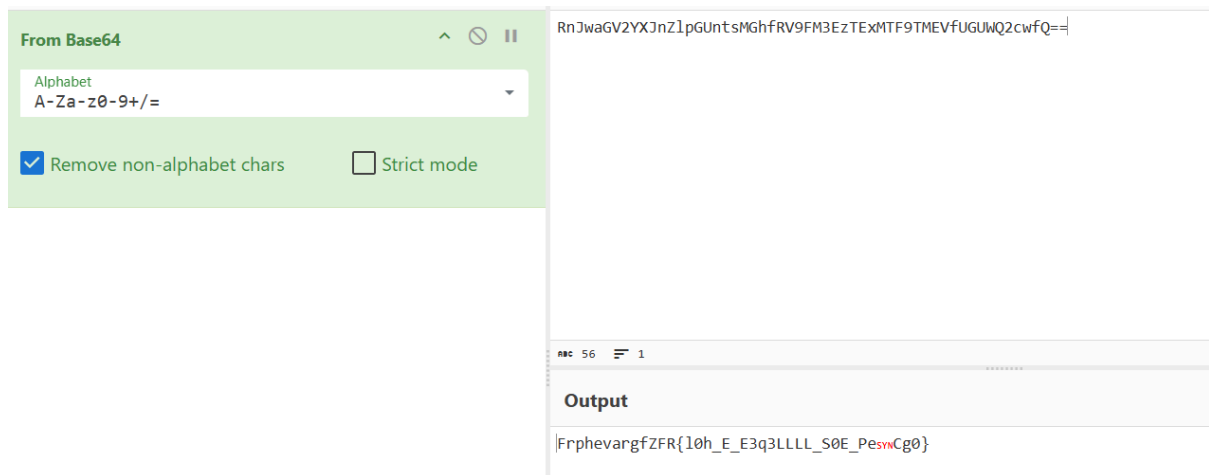
Hash functions examples:

- SHA-256
- SHA-3
- MD5 / SHA-1

Practice:

What is the flag?

RnJwaGV2YXJnZlpGUntsMGhfRV9FM3EzTExMTF9TMEVfUGUWQ2cwfQ==



The screenshot shows a web-based Base64 decoder. The input field contains the string "RnJwaGV2YXJnZlpGUntsMGhfRV9FM3EzTExMTF9TMEVfUGUWQ2cwfQ==". The output field displays the decoded string: "FrphevargfZFR{10h_E_E3q3LLLL_S0E_PesynCg0}". The interface includes a "From Base64" header, a dropdown menu for the alphabet (set to "A-Za-z0-9+/="), and checkboxes for "Remove non-alphabet chars" (checked) and "Strict mode" (unchecked).

From Base64

Alphabet
A-Za-z0-9+/=

☒ Remove non-alphabet chars ☐ Strict mode

RnJwaGV2YXJnZlpGUntsMGhfRV9FM3EzTExMTF9TMEVfUGUWQ2cwfQ==

Output

FrphevargfZFR{10h_E_E3q3LLLL_S0E_PesynCg0}

Recipe		Input
From Base64		RnJwaGV2YXJnZ1pGUntsMGhfRV9FM3EzTExMTF9TMEVfUGUwQ2cwfwQ==
Alphabet A-Za-z0-9+/=		
☒ Remove non-alphabet chars	☐ Strict mode	
ROT13		
☒ Rotate lower case chars	☒ Rotate upper case chars	
☐ Rotate numbers	Amount 13	
		Output
		SecurinetsMSE{y0u_R_R3d3YYYY_F0R_CrswPt0}

<https://www.dcode.fr/cipher-identifier>

<https://gchq.github.io/CyberChef/>

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